



# *Corporate Finance*

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# *What is Finance?*

# What is finance, in general?

- Finance studies how individuals and companies invest and value scarce resources over time and under uncertainty
- **Typical issues:** What is the fair value of an assets (e.g., a stock or bond), how should companies raise capital and invest it to improve their operations, how to evaluate risk?
- Applications:
  - Helps start-ups and mature businesses fund their operations and investments
  - Allows risk-averse individuals to save, invest, and finance purchases
  - Improves allocation of capital, resources, and risks in the economy, through markets and intermediaries
  - Goal is to promote overall prosperity, innovation, and growth

# Main finance topics

- **Corporate finance:** How companies manage their finances (assets and liabilities), raise capital for investments, evaluate investment decisions, etc.
- **Asset pricing:** How financial markets operate, how securities (e.g., stocks and bonds) are priced, how risk and return are priced in equilibrium, liquidity, trading strategies, derivatives, etc.
- **Household finance:** How individuals and households manage personal financial decisions, such as credit card debt, mortgages, credit scores, pensions, and other consumption/savings decisions

# Balance sheet view of company

- **Assets:** Resources
  - Operations and other sources of value
- **Liabilities:** Obligations
  - Where company has raised capital to fund investments and activities
  - Who owns and receives the value created by the company
- **Corporate finance:** How companies should manage assets and liabilities

| Assets                    | Liabilities    |
|---------------------------|----------------|
| Excess Cash ( $EC$ )      | Debt ( $D$ )   |
| Operating Assets ( $OA$ ) | Equity ( $E$ ) |

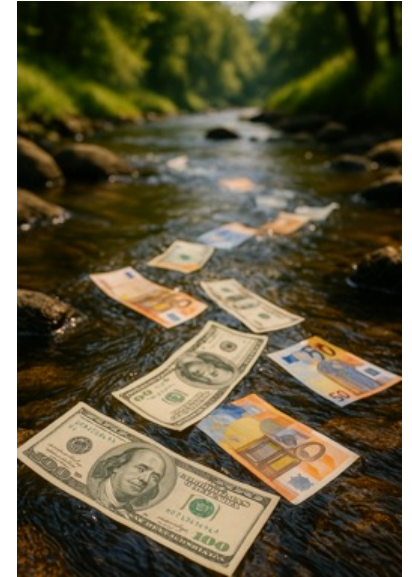
# *Cash Flows*



# Introducing: Cash flow (CF)

**Cash flow (CF)** is a fundamental concept in finance and economics

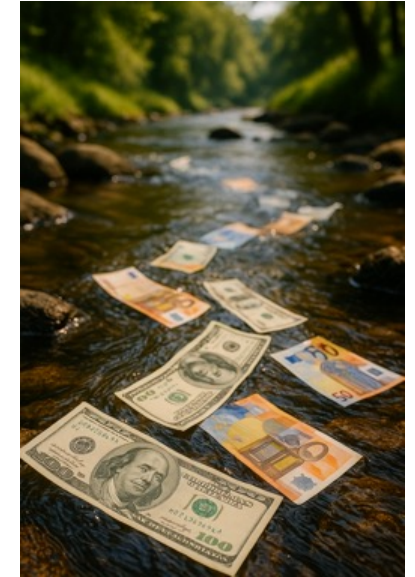
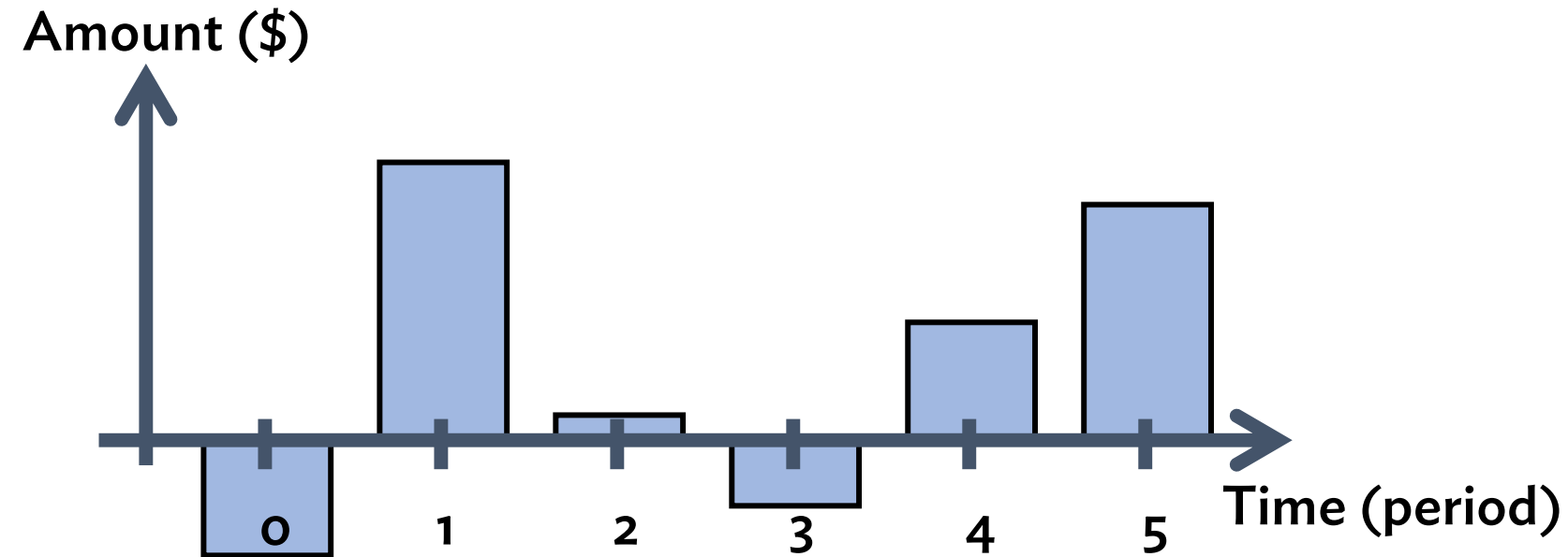
- It specifies money that is paid out and received over time
  - E.g., a company could be generating profits and paying suppliers



The value of a cash flow (CF) depends on three factors:

- **Amount:** A cash flow (CF) that gives more money is more valuable
- **Time-Value-of-Money:** A cash flow (CF) that pays sooner is more valuable
- **Risk:** A cash flow (CF) that is less risky is more valuable

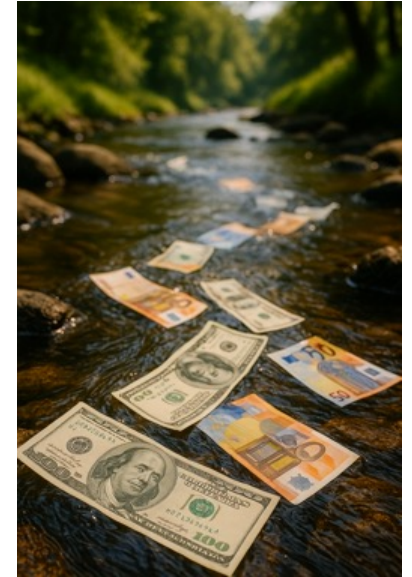
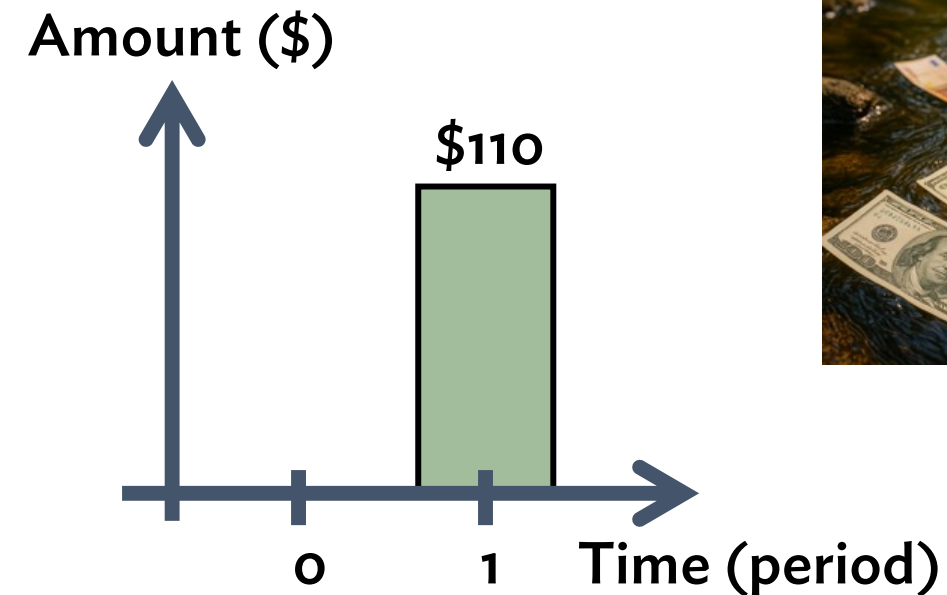
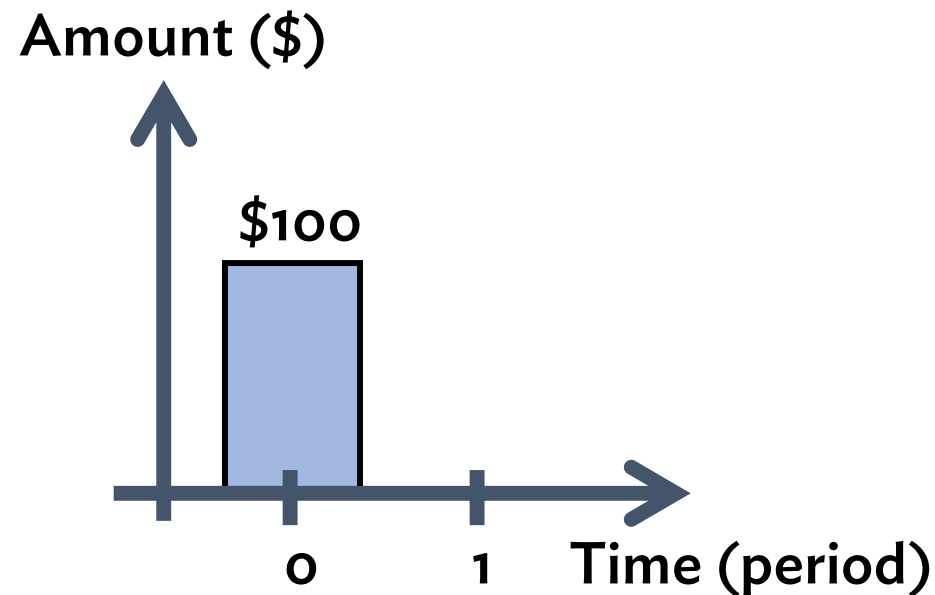
# A cash flow (CF)



- Positive bars are money coming in (received), negative bars are money going out (paid)
- A bar represents the (expected) amount at either a point in time or ongoing over the period
- **Conventions:** (1) The present moment (right now) is time zero; (2) ongoing cash flows are aggregated to end-of-period total amounts

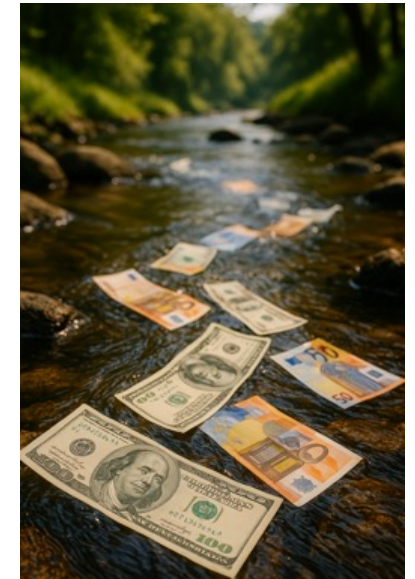
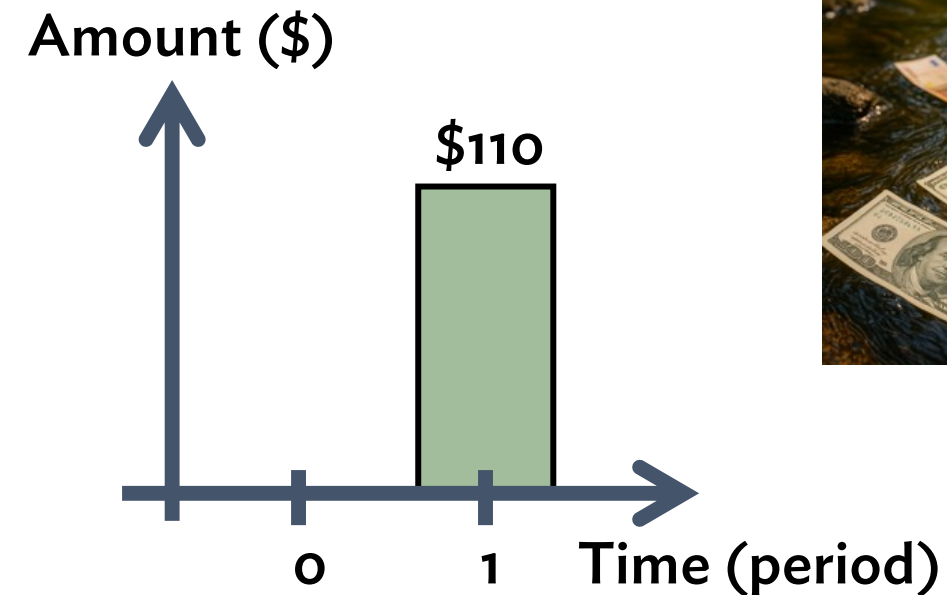
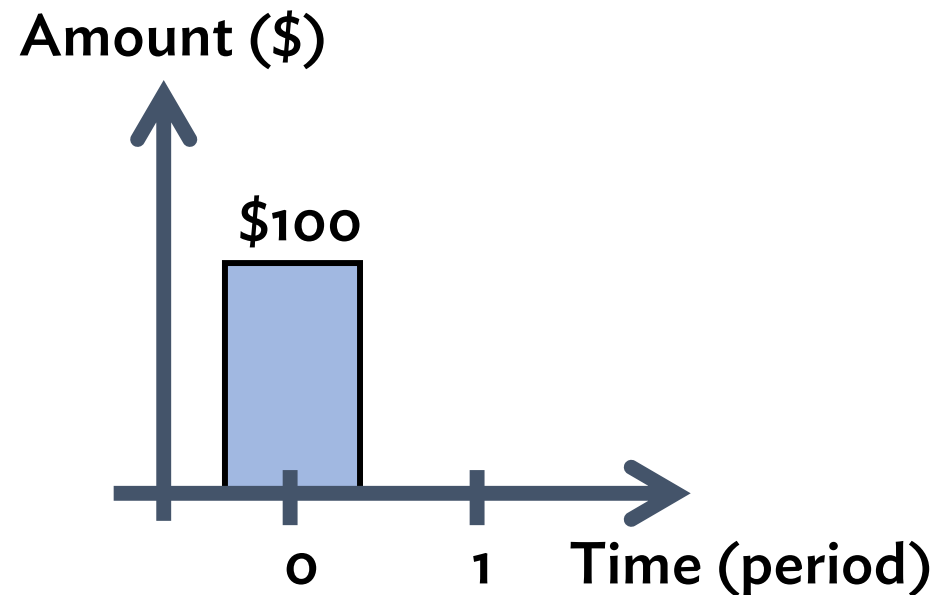


# Comparing two cash flows (CFs)



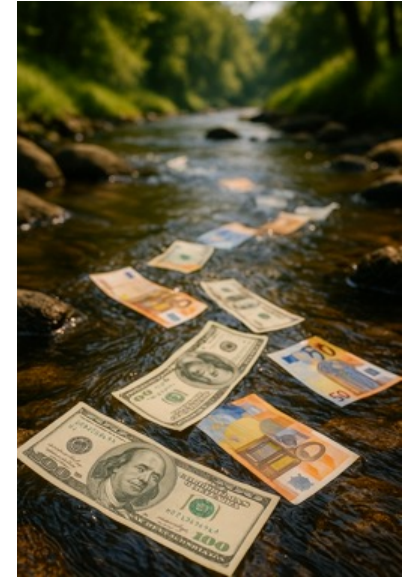
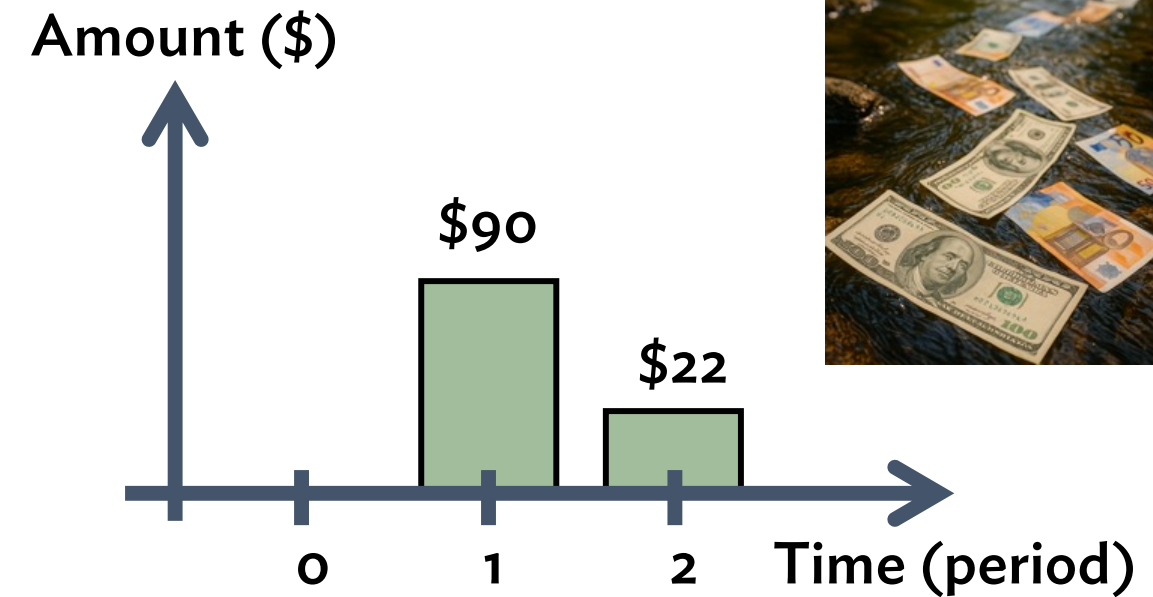
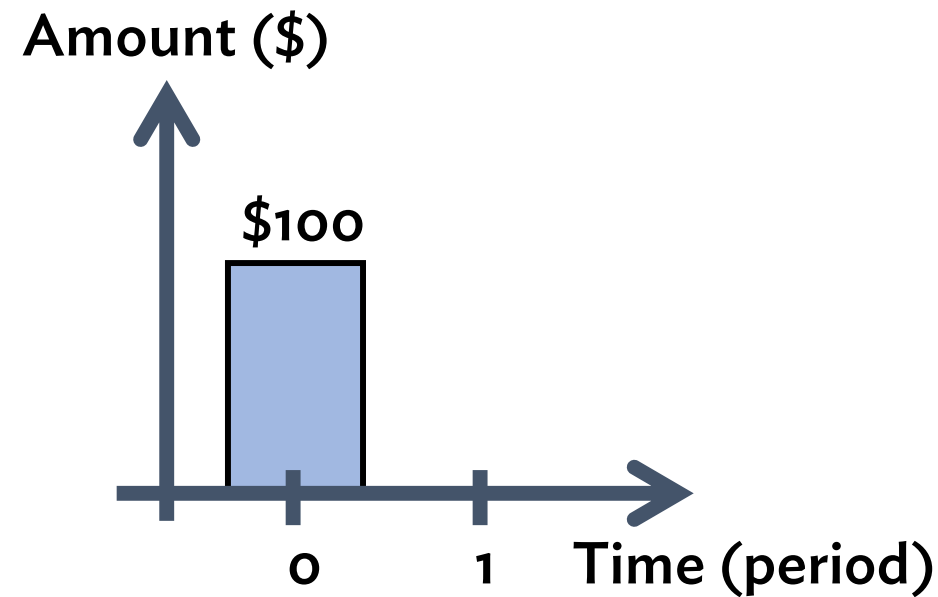
- Would you rather have \$100 today or \$110 next period?
  - Say we can freely borrow and lend at an interest rate of 10% per period
  - And, we want to see a movie tonight (time 0)

# Converting cash flows (CFs)



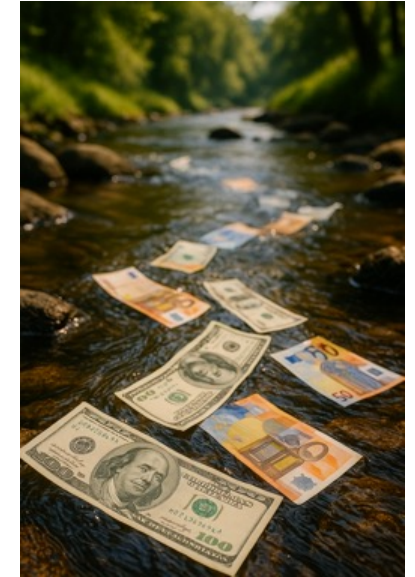
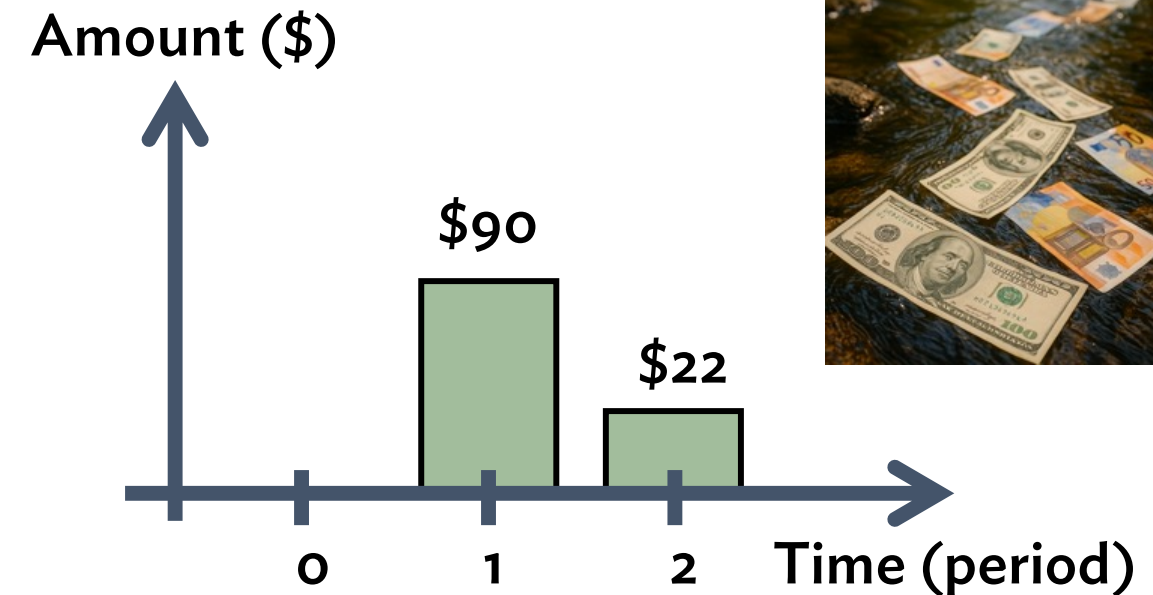
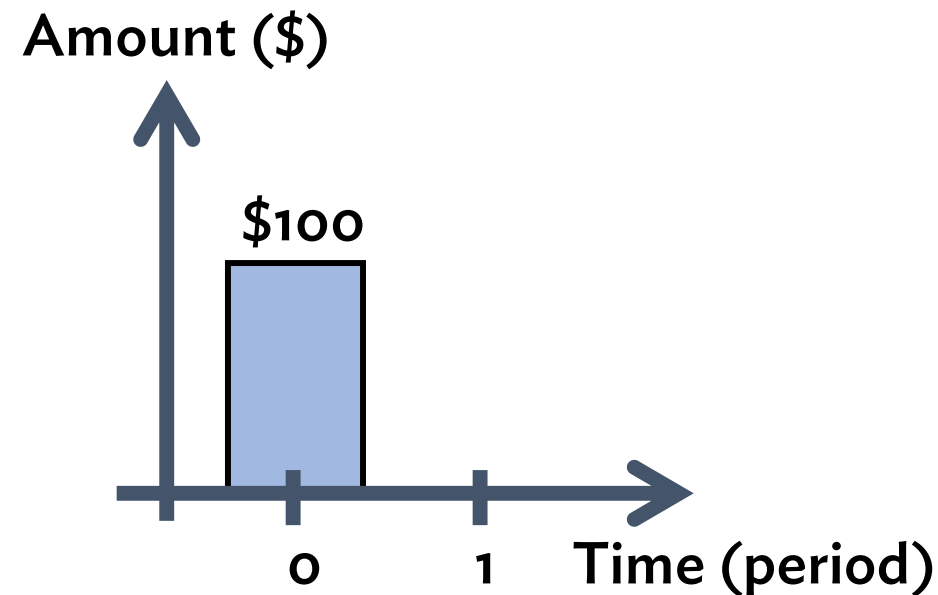
- Convert left to right CF by saving in bank [  $\$110 = \$100 \times (1+10\%)$  ]
- Convert right to left CF by borrowing from bank [  $\$100 = \$110 / (1+10\%)$  ] <- **IMPORTANT!**
- Hence, the two CFs are equally valuable

# More cash flows (CFs)



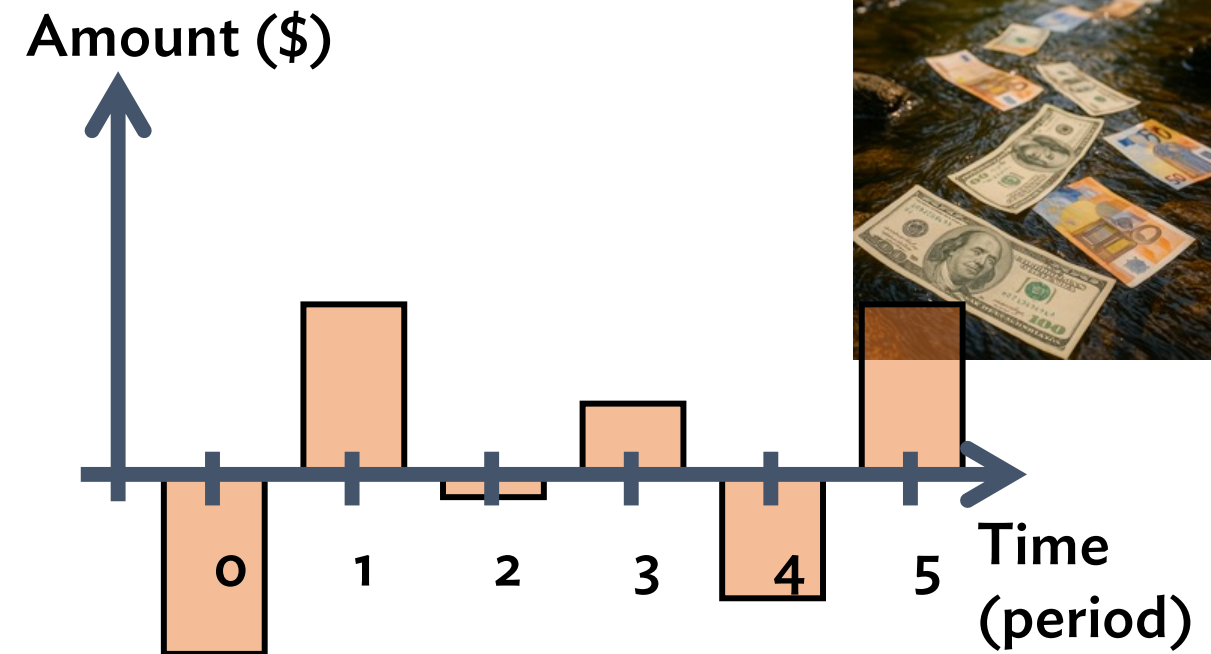
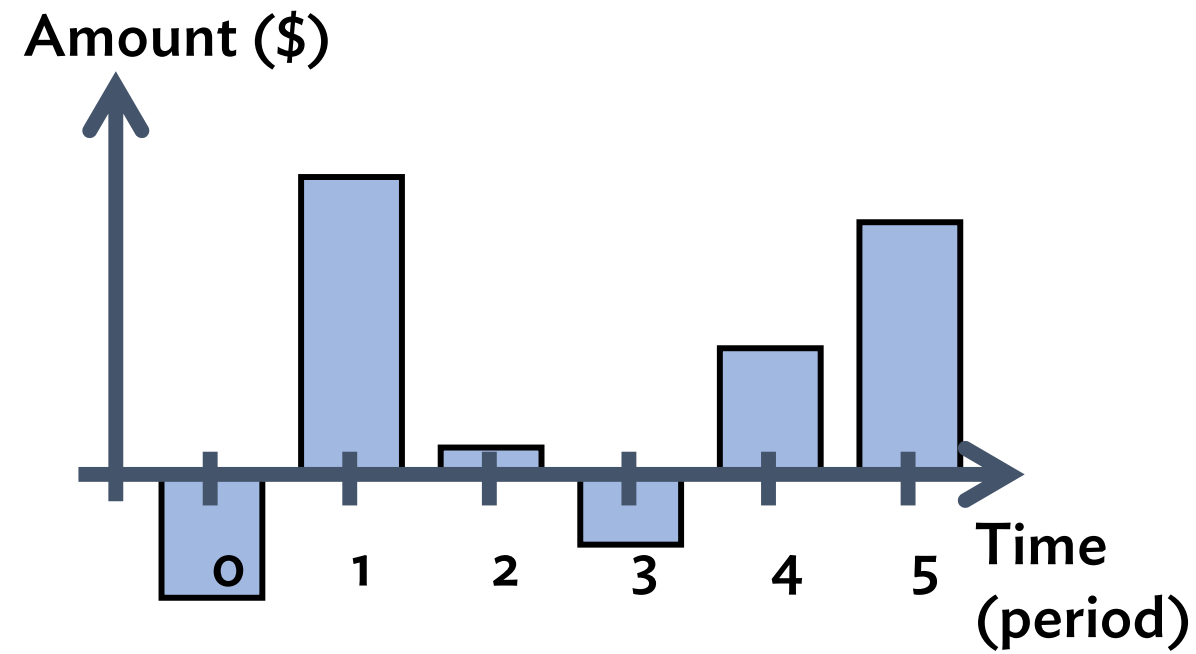
- Let's compare two (slightly) more complicated cash flows (CFs)
- Again, convert from left to the right CF by lending to bank
  - **Period 0:** Deposit \$100 in bank, which grows to \$110 at period 1
  - **Period 1:** Pay out \$90, and leave the remaining \$20 in bank, which grows to \$22

# More cash flows (CFs)



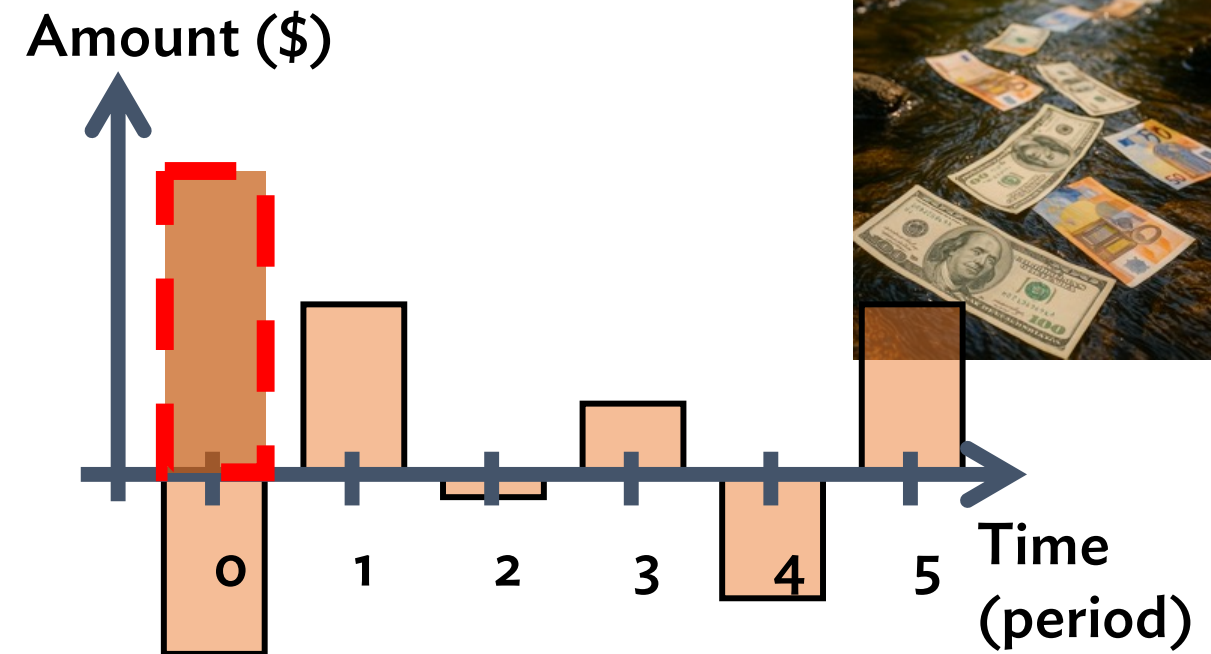
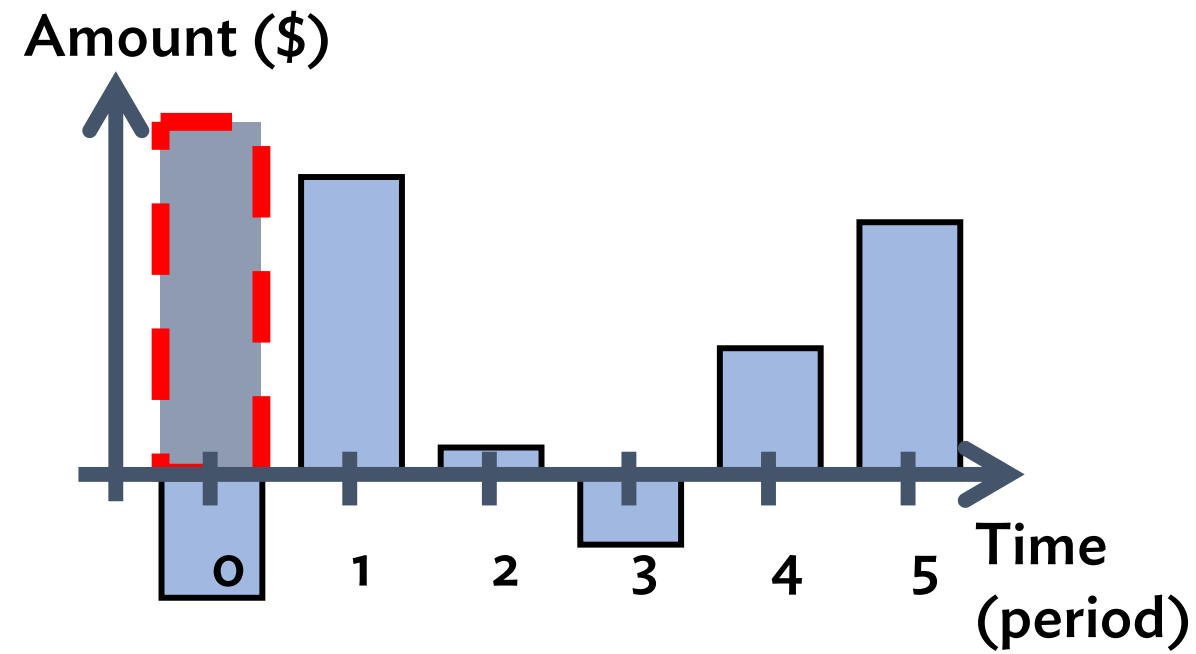
- Convert right CF to left one by borrowing from bank
  - **Period 0:** Borrow \$100 from bank
  - **Period 1:** We now owe bank \$110 [  $\$110 \times (1+10\%)$  ], repay \$90 with the money we receive, leaving outstanding balance of \$20 on our loan
  - **Period 2:** Repay remaining loan of \$22 [  $\$20 \times (1+10\%)$  ] with the money we receive
- Again, we can convert one to the other, and the two cash flows (CFs) are equivalent

# Complex cash flows (CFs)



- This looks complicated?!? How can we lend and borrow to turn one CF into the other?

# Complex cash flows (CFs)



- **Trick:** find amount “in the present” that is equivalent to each CF
- Present value (PV) is the amount (today) that can be converted into the CF, and vice versa



# Definition: Present value (PV)

- The present value (PV) of a cash flow (CF) is the amount today that is equivalent to the CF
  - If this amount were saved in bank (with interest), it could be turned into cash flow (CF)
  - Conversely, this amount could be borrowed from bank, when CF is used to repay loan (with interest)
- When comparing two cash flows (CFs), the one with the highest present value (PV) is always preferable
  - It can replicate the other CF, potentially with money left over
- When buying or selling a cash flow (CF), the present value (PV) is the fair price
  - PV is highest price you would pay (and lowest price you would accept) for CF

# Present value (PV) math

- Present Value ( $PV$ ) of a single future cash flow at time  $t$  ( $CF_t$ ) is:

$$PV = \frac{CF_t}{(1+r)^t}$$

Handwritten annotations: "Present value" above  $PV$ ; "cash flow" above  $CF_t$ ; "time" with an arrow pointing to  $t$  in the denominator; "discount rate" with an arrow pointing to  $r$  in the denominator.

- The per-period *discount rate* is  $r$  (e.g.,  $r = 10\%$ )
  - For a risk-free cash flow ( $CF$ ), discount rate is risk-free interest rate, i.e., rate that we can borrow and lend at (e.g., from bank)

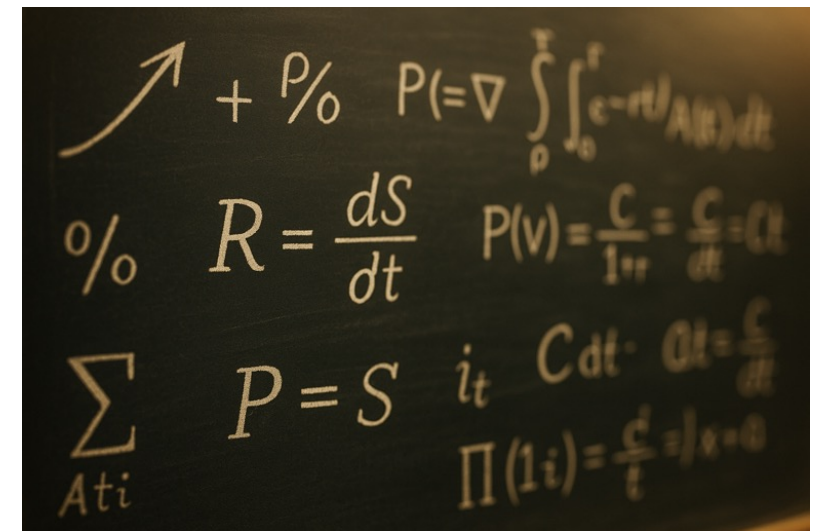


# Present value (PV) math

- Present Value (PV) of an entire cash flow stream

$$PV = \sum_{t=0}^T \frac{CF_t}{(1+r)^t}$$

- Cash flows (CFs) start today, at time  $t = 0$  (with the amount  $CF_0$ ) and end at terminal time  $T$  (with amount  $CF_T$ )
- PV of a stream of cash flows (CFs) is the sum of the PVs of each individual cash flow,  $CF_t$



# *Present Value (PV) in Excel*

# Brute-force present value (PV) calculation

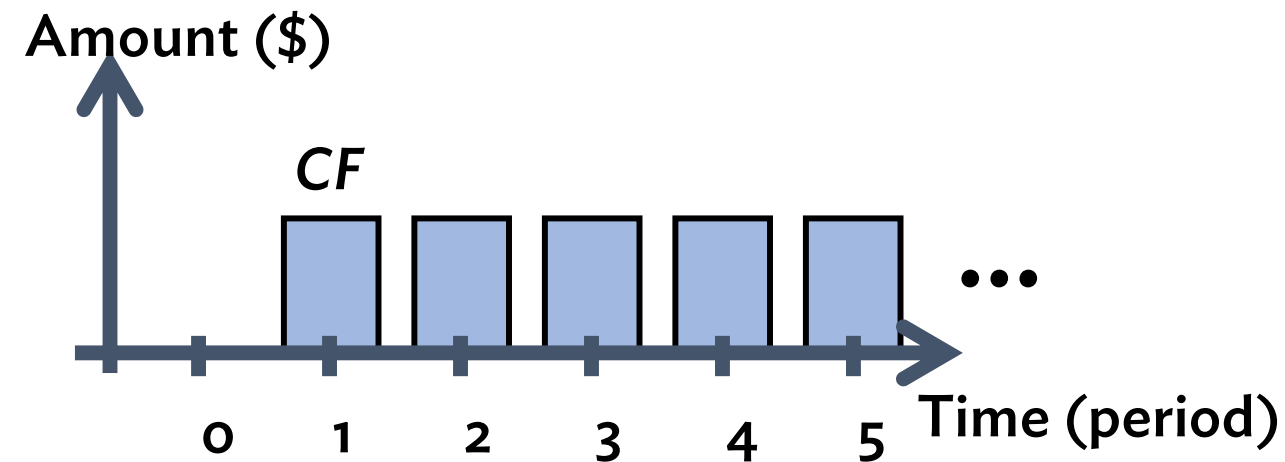
|   | A | B                         | C               | D                 | E        | F        | G               | H |
|---|---|---------------------------|-----------------|-------------------|----------|----------|-----------------|---|
| 1 |   |                           |                 |                   |          |          |                 |   |
| 2 |   | <b>Period</b>             | <b>0</b>        | <b>1</b>          | <b>2</b> | <b>3</b> |                 |   |
| 3 |   | Cash Flow (CF)            | -\$5.00         | \$3.00            | -\$2.00  | \$6.00   |                 |   |
| 4 |   | Discounted CF             | -\$5.00         | \$2.88            | -\$1.85  | \$5.33   | =F3/(1+\$C9)^F2 |   |
| 5 |   | <b>Present Value (PV)</b> | <b>\$1.3695</b> | =SUM(C4:G4)       |          |          |                 |   |
| 6 |   |                           |                 |                   |          |          |                 |   |
| 7 |   | <b>Present Value (PV)</b> | <b>\$1.3695</b> | =NPV(C9,D3:G3)+C3 |          |          |                 |   |
| 8 |   |                           |                 |                   |          |          |                 |   |
| 9 |   | <b>Discount Rate (r)</b>  | <b>4%</b>       |                   |          |          |                 |   |

# *Cash-Flow Math*

# Four special cash flows (CFs)

1. Perpetuity
2. Growing Perpetuity
3. Annuity
4. Growing Annuity

# 1. Present Value (PV) of perpetuity



- **Perpetuity:** A constant cash flow (CF) that starts next period ( $t = 1$ ) and continues forever (*in perpetuum*)
- Let's find its Present Value (PV)

# 1. Present Value (PV) of perpetuity

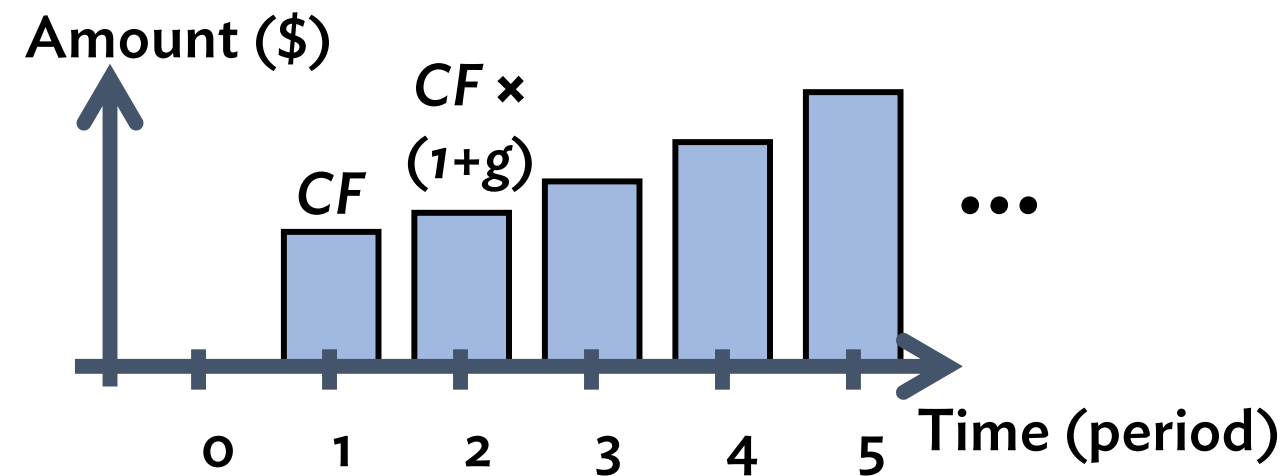
- Start from present value (PV) equation
  - **Note:**  $CF$  is constant in a perpetuity, so no subscript  $t$
- Multiply each side by  $(1+r)$
- Subtract first equation from second equation and rearrange
- Hence:  **$PV = CF/r$**
- **Intuition:** Annual interest on the amount  $PV$  is  $r \times PV = CF$
- **Math note:** This is an infinite geometric sequence, with multiplicative term  $[1/(1+r)]$

$$PV = \frac{CF}{(1+r)^1} + \frac{CF}{(1+r)^2} + \frac{CF}{(1+r)^3} + \dots$$

$$(1+r)PV = CF + \frac{CF}{(1+r)^1} + \frac{CF}{(1+r)^2} + \dots$$

$$rPV = CF \Rightarrow \boxed{PV = \frac{CF}{r}} = \frac{\$20}{10\%} = \$200$$

## 2. PV of growing perpetuity



- **Growing Perpetuity:** A cash flow ( $CF$ ) that starts next period ( $t=1$ ), with initial amount  $CF$ , continues forever, and grows by  $(1+g)$  each period
  - Cash flows,  $CF_1 = CF$ ,  $CF_2 = CF \times (1+g)$ ,  $CF_3 = CF \times (1+g)^2$ , *etc.*



## 2. PV of growing perpetuity

- Start from PV value equation
  - $CF$  is the first cash flow in the growing perpetuity
  - Cash flow grows by  $(1+g)$  each period

$$PV = \frac{CF}{(1+r)^1} + \frac{(1+g)CF}{(1+r)^2} + \frac{(1+g)^2 CF}{(1+r)^3} + \dots$$

- Multiply each side by  $(1+r)/(1+g)$
- Subtract first equation from second equation and rearrange

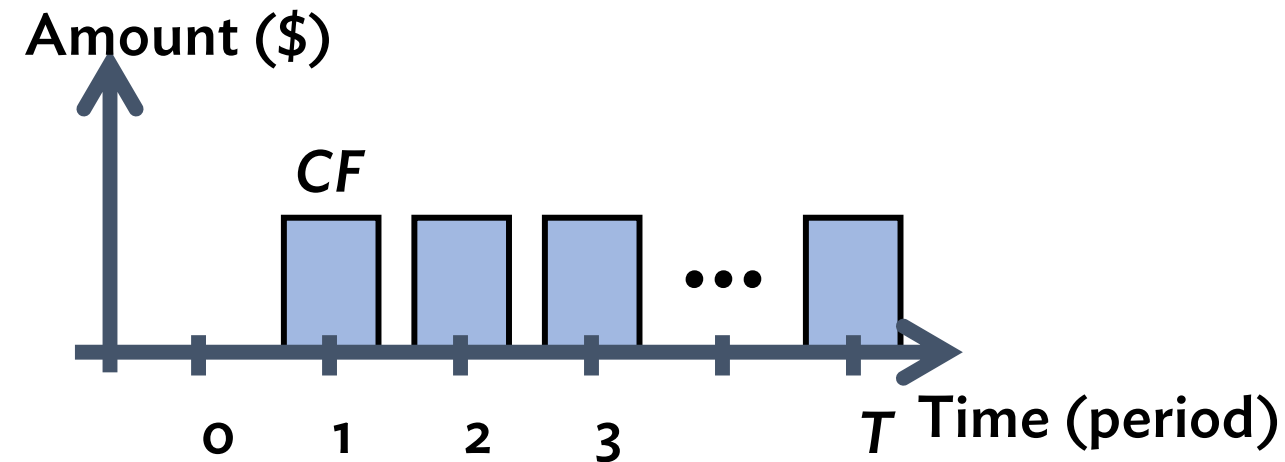
$$\frac{(1+r)}{(1+g)} PV = \frac{CF}{(1+g)} + \frac{CF}{(1+r)^1} + \frac{(1+g)CF}{(1+r)^2} + \dots$$

- Hence: **PV = CF/(r-g)**
- Note:** Math only works when  $r > g$ 
  - What happens when  $r < g$ ?

$$\left[ \frac{(1+r)}{(1+g)} - 1 \right] PV = \frac{CF}{(1+g)} \Rightarrow$$

$$PV = \frac{CF}{r - g}$$

### 3. PV of annuity



- **Annuity:** A constant cash flow ( $CF$ ) that starts next period ( $t = 1$ ) and ends at time  $T$

### 3. PV of annuity

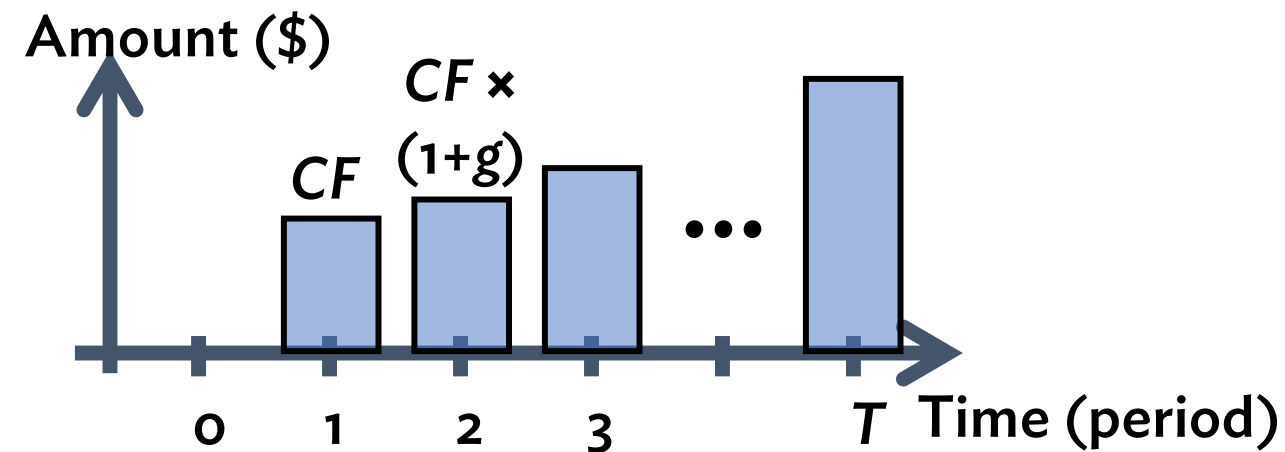
- Again, start from PV equation
- Multiply by  $(1+r)$
- Subtract and rearrange
- Note, it looks complicated, but it is just a factor times the initial cash flow (CF)
  - Like the other formulas

$$PV = \frac{CF}{(1+r)^1} + \frac{CF}{(1+r)^2} + \dots + \frac{CF}{(1+r)^T}$$

$$(1+r)PV = CF + \frac{CF}{(1+r)^1} + \dots + \frac{CF}{(1+r)^{T-1}}$$

$$rPV = CF - \frac{CF}{(1+r)^T} \Rightarrow PV = CF \frac{[1 - (1+r)^{-T}]}{r}$$

## 4. PV of growing annuity



- **Growing Annuity:** Cash flow that starts next period ( $t=1$ ) with amount  $CF$ , grows by  $(1+g)$  each period, and ends in period  $T$

# 4. PV of growing annuity

- Calculations as before

- **Note:** Unlike for growing perpetuity, this works for  $g > r$  (why?)

$$PV = \frac{CF}{(1+r)^1} + \frac{(1+g)CF}{(1+r)^2} + \dots + \frac{(1+g)^{T-1}CF}{(1+r)^T}$$

- But rarely used in practice

- With this complexity, it is usually better to brute-force Excel

$$\frac{(1+r)}{(1+g)} PV = \frac{CF}{(1+g)} + \frac{CF}{(1+r)^1} + \dots + \frac{(1+g)^{T-2}CF}{(1+r)^{T-1}}$$

$$\left[ \frac{(1+r)}{(1+g)} - 1 \right] PV = \frac{CF}{(1+g)} - \frac{(1+g)^{T-1}CF}{(1+r)^T} \Rightarrow PV = CF \frac{\left[ 1 - \left( \frac{1+r}{1+g} \right)^{-T} \right]}{r-g}$$

## 4. PV of growing annuity

|    | A | B                           | C          | D                                     | E        | F        | G        | H                 | I |
|----|---|-----------------------------|------------|---------------------------------------|----------|----------|----------|-------------------|---|
| 1  |   |                             |            |                                       |          |          |          |                   |   |
| 2  |   | <b>Period</b>               | <b>0</b>   | <b>1</b>                              | <b>2</b> | <b>3</b> | <b>4</b> |                   |   |
| 3  |   | Cash flow ( <i>CF</i> )     | \$0.00     | \$5.00                                | \$5.25   | \$5.51   | \$5.79   | =F3*(1+\$C7)      |   |
| 4  |   | Discounted CF               | \$0.00     | \$4.55                                | \$4.34   | \$4.14   | \$3.95   | =G3/(1+\$C8)^G\$2 |   |
| 5  |   | Present Value ( <i>PV</i> ) | \$16.97929 | =SUM(C4:G4)                           |          |          |          |                   |   |
| 6  |   |                             |            |                                       |          |          |          |                   |   |
| 7  |   | Growth Rate ( <i>g</i> )    | 5%         |                                       |          |          |          |                   |   |
| 8  |   | Discount rate ( <i>r</i> )  | 10%        |                                       |          |          |          |                   |   |
| 9  |   | Ending Period ( <i>T</i> )  | 4          |                                       |          |          |          |                   |   |
| 10 |   |                             |            |                                       |          |          |          |                   |   |
| 11 |   | Growing Annuity Formula     | \$16.97929 | =D3*(1-((1+C8)/(1+C7))^(−C9))/(C8−C7) |          |          |          |                   |   |

# Special cash flows (and their PVs)

## 1. Perpetuity:

- Start period 1

$$PV_0 = \frac{CF}{r}$$

## 2. Growing Perpetuity:

- Start period 1

$$PV_0 = \frac{CF}{r - g}$$

## 3. Annuity

- Start period 1, end period  $T$

$$PV_0 = CF \frac{[1 - (1 + r)^{-T}]}{r}$$

## 4. Growing Annuity

- Start period 1, end period  $T$

$$PV_0 = CF \frac{\left[1 - \left(\frac{1 + r}{1 + g}\right)^{-T}\right]}{r - g}$$